

We define the notions of biased and unbiased monoidal category in the enriched setting. We also define (representable) multicategories in this setting and show that each structure gives rise to the others. [need to consider maps and hopefully give equivalences of structures.](#)

Let $\mathcal{V}$ be a cartesian closed category, and let $\mathcal{C}$ be a $\mathcal{V}$ -category.

We utilise the following notational shorthand:

- For global elements of hom-objects, we write $f \in \mathcal{C}(A, B)$ or $f : A \rightarrow B$ to mean f has type $1 \rightarrow \mathcal{C}(A, B)$.
- For $g : B \rightarrow C$ and $h : Z \rightarrow A$ we denote by

$$g \circ - : \mathcal{C}(A, B) \rightarrow \mathcal{C}(A, C), \quad - \circ h : \mathcal{C}(A, B) \rightarrow \mathcal{C}(Z, B)$$

the following composites:

$$\begin{aligned} \mathcal{C}(A, B) &\cong 1 \times \mathcal{C}(A, B) \xrightarrow{g \times 1} \mathcal{C}(B, C) \times \mathcal{C}(A, B) \xrightarrow{\text{comp}} \mathcal{C}(A, C), \\ \mathcal{C}(A, B) &\cong \mathcal{C}(A, B) \times 1 \xrightarrow{1 \times h} \mathcal{C}(A, B) \times \mathcal{C}(Z, A) \xrightarrow{\text{comp}} \mathcal{C}(Z, B). \end{aligned}$$

For $g : B \rightarrow C$ and $h : A \rightarrow B$, the global element $g \circ h : A \rightarrow C$ is equal to both $(g \circ -)(h)$ and $(- \circ h)(g)$, both equalling the composite

$$1 \cong 1 \times 1 \xrightarrow{g \times h} \mathcal{C}(B, C) \times \mathcal{C}(A, B) \xrightarrow{\text{comp}} \mathcal{C}(A, C).$$

Since $\mathcal{V}$ is cartesian closed, it has finite products. Thus the product category $\mathcal{C}^n$ is also $\mathcal{V}$ -enriched, with

$$\mathcal{C}^n(\bar{X}, \bar{Y}) := \prod_{i=1}^n \mathcal{C}(X_i, Y_i).$$

Definition. A $\mathcal{V}$ -enriched unbiased monoidal category (or unbiased monoidal $\mathcal{V}$ -category) comprises:

- a $\mathcal{V}$ -category $\mathcal{C}$,
- for $n \geq 0$, $\mathcal{V}$ -functors $\bigotimes^n : \mathcal{C}^n \rightarrow \mathcal{C}$ such that $\bigotimes^1 = 1_{\mathcal{C}}$, and
- for $n \geq 0$ and $k_1, \dots, k_n \geq 0$, $\mathcal{V}$ -natural isomorphisms

$$\beta_{n, \bar{k}} : \bigotimes^n \circ \prod_{i=1}^n \bigotimes^{k_i} \rightarrow \bigotimes^k$$

(where $k = \sum_{i=1}^n k_i$) from

$$\bigotimes^n \circ \prod_{i=1}^n \bigotimes^{k_i} : \prod_{i=1}^n \mathcal{C}^{k_i} \rightarrow \mathcal{C}^n \rightarrow \mathcal{C}$$

to

$$\bigotimes^k : \mathcal{C}^k \rightarrow \mathcal{C},$$

such that $\beta_{n,\bar{1}} = 1_{\otimes^n}$ and such that the following diagram commutes for $k = \sum_{i=1}^n k_i$, $l_i = \sum_{j=1}^{k_i} l_{ij}$ and $l = \sum_{i=1}^n l_i$.

$$\begin{array}{ccc}
\otimes^n \circ \prod_i^n (\otimes^{k_i} \circ \prod_j^{k_i} \otimes^{l_{ij}}) & \xrightarrow{\otimes^n \circ \prod \beta_{k_i, \bar{l}_i}} & \otimes^n \circ \prod_i^n \otimes^{l_i} \\
\sim \downarrow & & \downarrow \beta_{n, \bar{l}} \\
\otimes^n \circ \prod_i^n \otimes^{k_i} \circ \prod_{i,j}^k \otimes^{l_{ij}} & & \\
\downarrow \beta_{n, \bar{k}} \circ \prod_{i,j}^k \otimes^{l_{ij}} & & \\
\otimes^k \circ \prod_{i,j}^k \otimes^{l_{ij}} & \xrightarrow{\beta_{k, \bar{l}}} & \otimes^l
\end{array}$$

For brevity, we may omit arity and composition symbols, and we furthermore denote Cartesian products by bars over symbols. Thus, we write

$$\beta : \otimes \bar{\otimes} \rightarrow \otimes$$

making the following diagram commute.

$$\begin{array}{ccc}
\otimes \bar{\otimes} \bar{\otimes} & \xrightarrow{\otimes \bar{\beta}} & \otimes \bar{\otimes} \\
\beta \bar{\otimes} \downarrow & & \downarrow \beta \\
\otimes \bar{\otimes} & \xrightarrow{\beta} & \otimes
\end{array}$$

Since $\otimes^n$ is a $\mathcal{V}$ -functor, it comes with maps

$$\mathcal{C}^n(\bar{X}, \bar{Y}) \xrightarrow{\otimes_{\bar{X}, \bar{Y}}^n} \mathcal{C}(\otimes \bar{X}, \otimes \bar{Y})$$

for all $\bar{X}, \bar{Y}$ in $\mathcal{C}$. These respect identities and composition, which is to say that $\otimes_{X, X}^1(1_X) = 1_{\otimes^1 X}$ and that the following diagram commutes.

$$\begin{array}{ccc}
\mathcal{C}^n(\bar{Y}, \bar{Z}) \times \mathcal{C}^n(\bar{X}, \bar{Y}) & \xrightarrow{\text{comp}} & \mathcal{C}^n(\bar{X}, \bar{Z}) \\
\otimes_{\bar{Y}, \bar{Z}}^n \times \otimes_{\bar{X}, \bar{Y}}^n \downarrow & & \downarrow \otimes_{\bar{X}, \bar{Z}}^n \\
\mathcal{C}(\otimes \bar{Y}, \otimes \bar{Z}) \times \mathcal{C}(\otimes \bar{X}, \otimes \bar{Y}) & \xrightarrow{\text{comp}} & \mathcal{C}(\otimes \bar{X}, \otimes \bar{Z})
\end{array}$$

Since β is a $\mathcal{V}$ -natural transformation, the following naturality square commutes.

$$\begin{array}{ccc}
\mathcal{C}^k(\bar{\bar{X}}, \bar{\bar{Y}}) & \xrightarrow{\otimes_{\bar{\bar{X}}, \bar{\bar{Y}}}} & \mathcal{C}^n(\otimes \bar{\bar{X}}, \otimes \bar{\bar{Y}}) \xrightarrow{\otimes_{\otimes \bar{\bar{X}}, \otimes \bar{\bar{Y}}}} \mathcal{C}(\otimes \otimes \bar{\bar{X}}, \otimes \otimes \bar{\bar{Y}}) \\
\otimes_{\bar{\bar{X}}, \bar{\bar{Y}}} \downarrow & & \downarrow \beta_{\bar{\bar{Y}}} \circ - \\
\mathcal{C}(\otimes \bar{\bar{X}}, \otimes \bar{\bar{Y}}) & \xrightarrow{- \circ \beta_{\bar{\bar{X}}}} & \mathcal{C}(\otimes \otimes \bar{\bar{X}}, \otimes \bar{\bar{Y}})
\end{array}$$

Definition. A $\mathcal{V}$ -enriched biased monoidal category (or biased monoidal $\mathcal{V}$ -category) comprises:

- a $\mathcal{V}$ -category $\mathcal{C}$,
- a unit $I \in \text{ob } \mathcal{C}$ and $\mathcal{V}$ -functor $- \otimes - : \mathcal{C}^2 \rightarrow \mathcal{C}$, and
- $\mathcal{V}$ -natural isomorphisms with components

$$\begin{aligned}\lambda_A &: I \otimes A \rightarrow A \\ \rho_A &: A \otimes I \rightarrow A \\ \alpha_{A,B,C} &: (A \otimes B) \otimes C \rightarrow A \otimes (B \otimes C)\end{aligned}$$

such that the two diagrams

$$\begin{array}{ccc} ((- \otimes -) \otimes -) \otimes - & \xrightarrow{\alpha} & (- \otimes -) \otimes (- \otimes -) \\ \alpha \downarrow & & \downarrow \alpha \\ (- \otimes (- \otimes -)) \otimes - & & \\ \alpha \downarrow & & \downarrow \alpha \\ - \otimes ((- \otimes -) \otimes -) & \xrightarrow{\alpha} & - \otimes (- \otimes (- \otimes -)) \end{array}$$

$$\begin{array}{ccc} (- \otimes I) \otimes - & \xrightarrow{\alpha} & - \otimes (I \otimes -) \\ & \searrow \rho & \downarrow \lambda \\ & & - \otimes - \end{array}$$

commute.

Proposition. *Every unbiased monoidal $\mathcal{V}$ -category is a biased monoidal $\mathcal{V}$ -category, and vice versa.*

Proof. Let $(\mathcal{C}, \otimes, \beta)$ be an unbiased monoidal $\mathcal{V}$ -category. We define

$$\begin{aligned}I &:= \overset{0}{\bigotimes}(*), \\ - \otimes - &:= \overset{2}{\bigotimes}, \\ \lambda &:= \beta_{2,(0,1)}, \\ \rho &:= \beta_{2,(1,0)}, \\ \alpha &:= \beta_{2,(1,2)}^{-1} \circ \beta_{2,(2,1)}.\end{aligned}$$

The following commuting diagrams verify the pentagon and triangle axioms,

using the single coherence axiom for β .

$$\begin{array}{ccccc}
((\cdot\cdot)\cdot) \cdot & \xrightarrow{\beta} & (\cdot\cdot)\cdot & \xrightarrow{\beta^{-1}} & (\cdot\cdot)(\cdot\cdot) \\
\beta \downarrow & & \beta \downarrow & & \downarrow \beta \\
(\cdot\cdot\cdot) \cdot & \xrightarrow{\beta} & \cdots & \xleftarrow{\beta} & \cdot\cdot(\cdot\cdot) \\
\beta^{-1} \downarrow & \nearrow \beta & & \nwarrow \beta & \downarrow \beta^{-1} \\
(\cdot(\cdot\cdot)) \cdot & \xrightarrow{\beta} \cdot(\cdot\cdot) \cdot & \xrightarrow{\beta^{-1}} \cdot((\cdot\cdot)\cdot) & \xrightarrow{\beta} \cdot(\cdot\cdot\cdot) & \xrightarrow{\beta^{-1}} \cdot(\cdot(\cdot\cdot))
\end{array}$$

$$\begin{array}{ccccc}
(\cdot I) \cdot & \xrightarrow{\beta} & \cdot I \cdot & \xrightarrow{\beta^{-1}} & \cdot(I \cdot) \\
\beta \downarrow & & \beta \downarrow & & \downarrow \beta \\
\cdot\cdot & \xrightarrow{\beta} & \cdot\cdot & \xleftarrow{\beta} & \cdot\cdot
\end{array}$$

In the other direction, let $(\mathcal{C}, I, \otimes, \lambda, \rho, \alpha)$ be a biased monoidal $\mathcal{V}$ -category. We define $\mathcal{V}$ -functors $\bigotimes^n$ as follows:

$$\begin{aligned}
\bigotimes^0 &: \mathcal{C}^0 \rightarrow \mathcal{C} : * \mapsto I, \\
\bigotimes^1 &: \mathcal{C}^1 \rightarrow \mathcal{C} := 1_{\mathcal{C}}, \\
\bigotimes^2 &: \mathcal{C}^2 \rightarrow \mathcal{C} := - \otimes -, \text{ and} \\
\bigotimes^n &: \mathcal{C}^n \rightarrow \mathcal{C} := - \otimes \bigotimes^{n-1} - \text{ for } n \geq 3.
\end{aligned}$$

For the $\mathcal{V}$ -natural isomorphisms $\beta_{n,\bar{k}}$ we use Mac Lane's Coherence Theorem for monoidal categories. This lets us define

$$\beta_{n,\bar{k}} : \bigotimes^n \circ \prod_{i=1}^n \bigotimes^{k_i} \rightarrow \bigotimes^k$$

as the unique arrow formed from the composition of λ, ρ and α maps mediating between the two bracketings. The same theorem further ensures that the β coherence diagram commutes. $\square$

Definition. A $\mathcal{V}$ -enriched multicategory (or $\mathcal{V}$ -multicategory) $\mathcal{C}$ comprises

- a collection of objects $\text{ob } \mathcal{C}$,
- for $n \geq 0$ and objects $\bar{X}, Y$ a hom-object $\mathcal{C}(\bar{X}; Y)$ in $\mathcal{V}$,
- for each object X an identity $1_X : X \rightarrow X$, and

- for $n \geq 0$, $k_1, \dots, k_n \geq 0$ and objects $\bar{\bar{X}}, \bar{Y}, Z$, a composition morphism in $\mathcal{V}$

$$\circ : \mathcal{C}(\bar{Y}; Z) \times \prod_{i=1}^n \mathcal{C}(\bar{X}_i; Y_i) \rightarrow \mathcal{C}(\bar{\bar{X}}; Z).$$

This data must satisfy associativity and unitality axioms expressed as commutative diagrams in $\mathcal{V}$.

- Associativity: for $n \geq 0$, $k_1, \dots, k_n \geq 0$, $l_1, \dots, l_{k_i} \geq 0$ and objects $\bar{\bar{W}}, \bar{\bar{X}}, \bar{Y}, Z$, the following diagram commutes.

$$\begin{array}{ccc} \mathcal{C}(\bar{Y}; Z) \times \prod_i^n \mathcal{C}(\bar{X}_i; Y_i) \times \prod_i^n \prod_j^{k_i} \mathcal{C}(\bar{W}_{ij}; X_{ij}) & \xrightarrow{\circ \times 1} & \mathcal{C}(\bar{\bar{X}}; Z) \times \prod_i^n \prod_j^{k_i} \mathcal{C}(\bar{W}_{ij}; X_{ij}) \\ \downarrow 1 \times \Pi \circ & & \downarrow \circ \\ \mathcal{C}(\bar{Y}; Z) \times \prod_i^n \mathcal{C}(\bar{\bar{W}}_i; Y_i) & \xrightarrow{\circ} & \mathcal{C}(\bar{\bar{W}}; Z) \end{array}$$

- Unitality: for $n \geq 0$ and objects $\bar{X}, Y$, the following diagrams commute.

$$\begin{array}{ccc} 1 \times \mathcal{C}(\bar{X}; Y) & \xrightarrow{1_Y \times 1} & \mathcal{C}(Y; Y) \times \mathcal{C}(\bar{X}; Y) \\ & \searrow \sim & \downarrow \circ \\ & & \mathcal{C}(\bar{X}; Y) \end{array}$$

$$\begin{array}{ccc} \mathcal{C}(\bar{X}; Y) \times 1 & \xrightarrow{1 \times \Pi 1_{X_i}} & \mathcal{C}(\bar{X}; Y) \times \prod_i^n \mathcal{C}(X_i; X_i) \\ & \searrow \sim & \downarrow \circ \\ & & \mathcal{C}(\bar{X}; Y) \end{array}$$

Remark. The objects and unary hom-objects of a $\mathcal{V}$ -enriched multicategory $\mathcal{C}$, with the inherited identities and composition, form a $\mathcal{V}$ -category which we call $\mathcal{C}_1$.

Definition. An arrow

$$u : \bar{X} \rightarrow Y$$

is called *universal* if precomposition with u defines an isomorphism

$$- \circ u : \mathcal{C}(Y; Z) \xrightarrow{\sim} \mathcal{C}(\bar{X}; Y)$$

natural in Z .

Remark. If $u : \bar{X} \rightarrow Y$ and $v : \bar{X} \rightarrow Z$ are both universal arrows for the same list of objects, then there is a unique isomorphism $a : Y \rightarrow Z$ with $a \circ u = v$, given by the Yoneda lemma applied to the composite

$$\mathcal{C}(Z; W) \xrightarrow{- \circ v} \mathcal{C}(\bar{X}; W) \xrightarrow{(- \circ u)^{-1}} \mathcal{C}(Y; W).$$

Definition. A $\mathcal{V}$ -multicategory is called *representable* if

1. for $n \geq 0$ and objects $\bar{X}$ in $\mathcal{C}$, there is an object $\bigotimes^n \bar{X}$ and a universal arrow $u_{\bar{X}} : 1 \rightarrow \mathcal{C}(\bar{X}; \bigotimes^n \bar{X})$, and
2. universal arrows are closed under composition, which is to say that for $n \geq 0, k_1, \dots, k_n \geq 0$ and objects $\bar{X}$, the composite

$$u_{\prod \bigotimes \bar{X}} \circ \prod_i u_{\bar{X}_i} : \bar{X} \rightarrow \bigotimes^n \prod \bigotimes^{k_i} \bar{X}$$

is universal for $\bar{X}$.

Proposition. Every representable $\mathcal{V}$ -multicategory can be given the structure of an unbiased monoidal $\mathcal{V}$ -category, and vice versa.

Proof. ($\Rightarrow$) Suppose $\mathcal{C}$ is a representable $\mathcal{V}$ -multicategory. We construct an unbiased monoidal $\mathcal{V}$ -category structure on the $\mathcal{V}$ -category $\mathcal{C}_1$. For $n \geq 0$ we define $\mathcal{V}$ -functors $\bigotimes^n : \mathcal{C}^n \rightarrow \mathcal{C}$ whose action on objects is determined by representability. On morphisms, we need to construct maps of shape

$$\mathcal{C}_1^n(\bar{X}, \bar{Y}) \rightarrow \mathcal{C}_1(\bigotimes^n \bar{X}; \bigotimes^n \bar{Y});$$

these are given by the composites

$$\prod_i^n \mathcal{C}(X_i, Y_i) \xrightarrow{u_{\bar{Y}} \circ -} \mathcal{C}(\bar{X}; \bigotimes^n \bar{Y}) \xrightarrow{(- \circ u_{\bar{X}})^{-1}} \mathcal{C}(\bigotimes^n \bar{X}; \bigotimes^n \bar{Y}).$$

This preserves identities due to the unitality of $\circ$ in $\mathcal{C}$; it preserves composition due to the associativity of $\circ$ in $\mathcal{C}$.

$$\begin{array}{ccc} \prod_i^n 1 & \xrightarrow{\prod 1_{X_i}} & \prod_i^n \mathcal{C}(X_i, X_i) \\ \downarrow \sim & & \downarrow u \circ - \\ & \nearrow u & \mathcal{C}(\bar{X}; \bigotimes^n \bar{X}) \\ 1 & \xrightarrow{1_{\bigotimes \bar{X}}} & \mathcal{C}(\bigotimes^n \bar{X}; \bigotimes^n \bar{X}) \end{array} \quad \begin{array}{c} \uparrow - \circ u \end{array}$$

$$\begin{array}{ccc}
\prod_i^n \mathcal{C}(Y_i; Z_i) \times \prod_i^n \mathcal{C}(X_i; Y_i) & \xrightarrow{\Pi \circ} & \prod_i^n \mathcal{C}(X_i; Z_i) \\
(u \circ -) \times 1 \downarrow & & \downarrow u \circ - \\
\mathcal{C}(\bar{Y}; \otimes \bar{Z}) \times \prod_i^n \mathcal{C}(X_i; Y_i) & \xrightarrow{\circ} & \mathcal{C}(\bar{X}; \otimes \bar{Z}) \\
(- \circ u) \times 1 \uparrow & & \parallel \\
\mathcal{C}(\otimes \bar{Y}; \otimes \bar{Z}) \times \prod_i^n \mathcal{C}(X_i; Y_i) & & \\
1 \times (u \circ -) \downarrow & & \\
\mathcal{C}(\otimes \bar{Y}; \otimes \bar{Z}) \times \mathcal{C}(\bar{X}; \otimes \bar{Y}) & \xrightarrow{\circ} & \mathcal{C}(\bar{X}; \otimes \bar{Z}) \\
1 \times (- \circ u) \uparrow & & \uparrow - \circ u \\
\mathcal{C}(\otimes \bar{Y}; \otimes \bar{Z}) \times \mathcal{C}(\otimes \bar{X}; \otimes \bar{Y}) & \xrightarrow{\circ} & \mathcal{C}(\otimes \bar{X}; \otimes \bar{Z})
\end{array}$$

So have have the required $\mathcal{V}$ -functors $\otimes^n : \mathcal{C}_1^n \rightarrow \mathcal{C}_1$. Now we construct $\mathcal{V}$ -natural isomorphisms

$$\beta_{n, \bar{k}} : \otimes^n \prod \otimes^{k_i} \Rightarrow \otimes^k$$

whose components are the isomorphisms given by the fact that there are universal arrows from $\bar{X}$ into both $\otimes^n \prod \otimes^{k_i}$ and $\otimes^k$.

It remains to show that the β satisfy the coherence equation. Let $\bar{\bar{X}}$ be a triply indexed sequence of objects. The coherence diagram for β relates four different bracketings of $\otimes X$. Let the corresponding tensor product objects (from the four corners of the diagram) be X_A, X_B, X_C and X_D . By the ‘closure under composition’ axiom, each of these has a universal arrow from $\bar{\bar{X}}$, u_A, u_B, u_C and u_D respectively, formed by composing elementary universal arrows using $\circ$. Writing β_{AB} for the isomorphism from $X_A \rightarrow X_B$, and so on, we need to show that $\beta_{BD} \circ \beta_{AB} = \beta_{CD} \circ \beta_{AC}$.

$$\begin{array}{ccc}
X_A & \xrightarrow{\beta_{AB}} & X_B \\
\beta_{AC} \downarrow & & \downarrow \beta_{BD} \\
X_C & \xrightarrow{\beta_{CD}} & X_D
\end{array}$$

By construction we have $\beta_{PQ} \circ u_P = u_Q$ for any bracketings P and Q . But then

$$\beta_{BD} \circ \beta_{AB} \circ u_A = \beta_{BD} \circ u_B = u_D = \beta_{CD} \circ u_C = \beta_{CD} \circ \beta_{AC} \circ u_A,$$

and so by universality of u_A the required coherence condition holds, and hence every representable $\mathcal{V}$ -multicategory is an unbiased monoidal $\mathcal{V}$ -category.

($\Leftarrow$) Suppose $\mathcal{C}$ is an unbiased monoidal $\mathcal{V}$ -category. We construct a representable $\mathcal{V}$ -multicategory $\mathcal{C}_M$ on the same objects and identities. For $n \geq 0$ the

hom-objects $\mathcal{C}_M(\bar{X}; Y)$ are defined by

$$\mathcal{C}(\bigotimes^n \bar{X}; Y).$$

Composition

$$\mathcal{C}_M(\bar{Y}; Z) \times \prod_{i=1}^n \mathcal{C}_M(\bar{X}_i; Y_i) \rightarrow \mathcal{C}_M(\bar{X}; Z)$$

is defined by the composite

$$\begin{aligned} \mathcal{C}(\bigotimes^n \bar{Y}, Z) \times \prod_{i=1}^n \mathcal{C}(\bigotimes^{k_i} \bar{X}_i, Y_i) &\xrightarrow{1 \times \bigotimes^n} \mathcal{C}(\bigotimes^n Y, Z) \times \mathcal{C}(\bigotimes^n \prod \bigotimes^{k_i} \bar{X}_i, \bigotimes^n \bar{Y}) \\ &\xrightarrow{\circ} \mathcal{C}(\bigotimes^n \prod \bigotimes^{k_i} \bar{X}_i, Z) \xrightarrow{- \circ \beta^{-1}} \mathcal{C}(\bigotimes^k \bar{X}, Z). \end{aligned}$$

This composition is associative and unital due to the coherence axiom for β and the identity conditions on β , respectively. Thus, $\mathcal{C}_M$ is a $\mathcal{V}$ -multicategory.

To show $\mathcal{C}_M$ is representable:

1. For any list of objects $\bar{X}$, we choose the object $\bigotimes^n \bar{X}$ and define the universal arrow $u_{\bar{X}} : \bar{X} \rightarrow \bigotimes^n \bar{X}$ to be the identity morphism $1_{\bigotimes^n \bar{X}} \in \mathcal{C}(\bigotimes^n \bar{X}, \bigotimes^n \bar{X})$. Pre-composition with $u_{\bar{X}}$ induces an isomorphism

$$\mathcal{C}_M(\bigotimes^n \bar{X}; Y) \xrightarrow{\sim} \mathcal{C}_M(\bar{X}; Y)$$

and so $u_{\bar{X}}$ is universal.

2. To check that universal arrows are closed under composition, we must show that the composite

$$v_{\bar{X}} := u_{\prod \bigotimes \bar{X}} \circ \prod_i u_{\bar{X}_i} : \bar{X} \rightarrow \bigotimes^n \prod \bigotimes^{k_i} \bar{X}_i$$

is universal for $\bar{X}$. This composite arrow $v_{\bar{X}}$ corresponds to the morphism $\beta_{n, \bar{k}}^{-1} : \bigotimes^k \bar{X} \rightarrow \bigotimes^n \prod \bigotimes^{k_i} \bar{X}_i$ in $\mathcal{C}$. Since $\beta_{n, \bar{k}}$ is a natural isomorphism, the map $- \circ (\beta_{n, \bar{k}})^{-1}_{\bar{X}}$ is an isomorphism, which confirms that $v_{\bar{X}}$ is a universal arrow.

Thus, $\mathcal{C}_M$ is a representable $\mathcal{V}$ -multicategory. $\square$

1 2-categories

1.1 uMonCat

Definition. Let $\mathcal{C}$ and $\mathcal{D}$ be unbiased monoidal $\mathcal{V}$ -categories. A *lax monoidal $\mathcal{V}$ -functor* between them comprises a $\mathcal{V}$ -functor $F : \mathcal{C} \rightarrow \mathcal{D}$ along with $\mathcal{V}$ -natural

transformations

$$\mu_n : \bigotimes^n F \rightarrow F \bigotimes^n : \mathcal{C}^n \rightarrow \mathcal{D}$$

for $n \geq 0$, such that $\mu_1 = 1_F$, and satisfying the coherence condition that the following diagram commutes:

$$\begin{array}{ccc} \bigotimes^n \prod \bigotimes^{k_i} F^k & \xrightarrow{\beta_{n,\bar{k}}} & \bigotimes^k F^k \\ \bigotimes^n \prod \mu_{k_i} \downarrow & & \downarrow \mu_k \\ \bigotimes^n F^n \prod \bigotimes^{k_i} & & \\ \mu_n \downarrow & & \\ F \bigotimes^n \prod \bigotimes^{k_i} & \xrightarrow{F\beta_{n,\bar{k}}} & F \bigotimes^k \end{array}$$

Definition. Let $(F, \mu), (G, \nu) : \mathcal{C} \rightarrow \mathcal{D}$ be lax monoidal functors between unbiased monoidal $\mathcal{V}$ -categories. A *monoidal V -natural transformation* between them comprises a V -natural transformation $\gamma : F \Rightarrow G$ between the underlying functors for which the following diagram commutes for all $n \geq 0$:

$$\begin{array}{ccc} \bigotimes^n F^n & \xrightarrow{\bigotimes^n \gamma^n} & \bigotimes^n G^n \\ \mu_n \downarrow & & \downarrow \nu_n \\ F \bigotimes^n & \xrightarrow{\gamma_{\otimes}} & G \bigotimes^n \end{array}$$

Proposition. *There is a (strict) 2-category $\mathcal{V}\text{-uMonCat}$ of unbiased monoidal $\mathcal{V}$ -categories, lax monoidal $\mathcal{V}$ -functors and monoidal $\mathcal{V}$ -natural transformations.*

Proof. Let $(F, \mu_F) : \mathcal{C} \rightarrow \mathcal{D}$ and $(G, \mu_G) : \mathcal{D} \rightarrow \mathcal{E}$ be lax monoidal functors between unbiased monoidal $\mathcal{V}$ -categories. We define

$$\mu_{GF,n} : \bigotimes^n (GF)^n \rightarrow GF \bigotimes^n : \mathcal{C}^n \rightarrow \mathcal{E}$$

by the composite

$$(G\mu_{F,n}) \circ (\mu_{G,n} F^n) : \bigotimes^n G^n F^n \rightarrow G \bigotimes^n F^n \rightarrow GF \bigotimes^n.$$

This satisfies the coherence condition since μ_F and μ_G do so:

$$\begin{array}{ccc}
\bigotimes^n \prod \bigotimes^{k_i} G^k F^k & \xrightarrow{\beta_{n,\bar{k}}} & \bigotimes^k G^k F^k \\
\downarrow \bigotimes \prod \mu_{G,k_i} & & \downarrow \mu_{G,k} F^k \\
\bigotimes^n G^n \prod \bigotimes^{k_i} F^k & \xrightarrow{\mu_{G,n}} G \bigotimes^n \prod \bigotimes^{k_i} F^k \xrightarrow{G\beta_{n,\bar{k}}} G \bigotimes^k F^k \\
\downarrow \bigotimes \prod \mu_{F,k_i} & \downarrow \bigotimes \prod \mu_{F,k_i} & \downarrow G\mu_{F,k} \\
\bigotimes^n G^n F^n \prod \bigotimes^{k_i} & \xrightarrow{\mu_{G,n}} G \bigotimes^n F^n \prod \bigotimes^{k_i} & \\
& \downarrow \mu_{F,n} & \\
& GF \bigotimes^n \prod \bigotimes^{k_i} & \xrightarrow{GF\beta_{n,\bar{k}}} GF \bigotimes^k
\end{array}$$

Hence this definition gives a lax monoidal $\mathcal{V}$ -functor (GF, μ_{GF}) . The composition is unital with identity $(1, 1) : \mathcal{C} \rightarrow \mathcal{C}$, and it is associative:

$$\begin{aligned}
\mu_{(HG)F,n} &:= (HG\mu_{F,n}) \circ (\mu_{HG,n} F^n) \\
&= (HG\mu_{F,n}) \circ (H\mu_{G,n} F^n) \circ (\mu_{H,n} G^n F^n) \\
&= (H\mu_{GF,n}) \circ (\mu_{H,n} (GF)^n) \\
&= \mu_{H(GF),n}.
\end{aligned}$$

The composite of two monoidal $\mathcal{V}$ -natural transformations is again monoidal, since the following diagram commutes:

$$\begin{array}{ccccc}
\bigotimes^n F^n & \xrightarrow{\bigotimes^n \gamma^n} & \bigotimes^n G^n & \xrightarrow{\bigotimes^n \delta^n} & \bigotimes^n H^n \\
\downarrow \mu_{F,n} & & \downarrow \mu_{G,n} & & \downarrow \mu_{H,n} \\
F \bigotimes^n & \xrightarrow{\gamma_\otimes} & G \bigotimes^n & \xrightarrow{\delta_\otimes} & H \bigotimes^n
\end{array}$$

Associativity and unitality are inherited from $\mathcal{V}\text{-Cat}$, since these are merely $\mathcal{V}$ -natural transformations satisfying a given property.

Hence we obtain the desired strict 2-category $\mathcal{V}\text{-uMonCat}$. $\square$

1.2 MonCat

Definition. Let $\mathcal{C}$ and $\mathcal{D}$ be biased monoidal $\mathcal{V}$ -categories. A *lax monoidal $\mathcal{V}$ -functor* between them comprises a $\mathcal{V}$ -functor $F : \mathcal{C} \rightarrow \mathcal{D}$ along with maps

$$\phi : I \rightarrow FI, \quad \phi_{X,Y} : FX \otimes FY \rightarrow F(X \otimes Y)$$

for X, Y in $\mathcal{C}$, making the followings coherence diagrams commute:

$$\begin{array}{ccc}
FI \otimes FX & \xrightarrow{\phi} & F(I \otimes X) \\
\uparrow \phi \otimes 1 & & \downarrow F\lambda \\
I \otimes FX & \xrightarrow{\lambda} & FX
\end{array}
\quad
\begin{array}{ccc}
FX \otimes FI & \xrightarrow{\phi} & F(X \otimes I) \\
\uparrow 1 \otimes \phi & & \downarrow F\rho \\
FX \otimes I & \xrightarrow{\rho} & FX
\end{array}$$

$$\begin{array}{ccccc}
(FA \otimes FB) \otimes FC & \xrightarrow{\phi \otimes 1} & F(A \otimes B) \otimes FC & \xrightarrow{\phi} & F((A \otimes B) \otimes C) \\
\alpha \downarrow & & & & \downarrow F\alpha \\
FA \otimes (FB \otimes FC) & \xrightarrow{1 \otimes \phi} & FA \otimes F(B \otimes C) & \xrightarrow{\phi} & F(A \otimes (B \otimes C))
\end{array}$$

Definition. Let $(F, \phi), (G, \psi) : \mathcal{C} \rightarrow \mathcal{D}$ be lax monoidal V -functors between biased monoidal $\mathcal{V}$ -categories. A *monoidal V -natural transformation* between them comprises a V -natural transformation $\gamma : F \Rightarrow G$ between the underlying functors for which the following diagrams commute for all X, Y in $\mathcal{C}$:

$$\begin{array}{ccc}
I & \xrightarrow{\phi} & FI \\
& \searrow \psi & \downarrow \gamma_I \\
& & GI
\end{array}
\quad
\begin{array}{ccc}
FA \otimes FB & \xrightarrow{\phi} & F(A \otimes B) \\
\gamma_A \otimes \gamma_B \downarrow & & \downarrow \gamma_{A \otimes B} \\
GA \otimes GB & \xrightarrow{\psi} & G(A \otimes B)
\end{array}$$

Proposition. *There is a (strict) 2-category $\mathcal{V}\text{-MonCat}$ of biased monoidal $\mathcal{V}$ -categories, lax monoidal $\mathcal{V}$ -functors and monoidal $\mathcal{V}$ -natural transformations.*

1.3 MultiCat

Definition. Let $\mathcal{C}$ and $\mathcal{D}$ be $\mathcal{V}$ -multicategories. A $\mathcal{V}$ -multifunctor between them comprises maps of hom-objects

$$F_{\bar{X}, Y} : \mathcal{C}(\bar{X}; Y) \rightarrow \mathcal{D}(F\bar{X}; FY)$$

for $\bar{X}, Y$ in $\mathcal{C}$, such that $F_{X, X}(1_X) = 1_{FX}$ and the following coherence diagram commutes:

$$\begin{array}{ccc}
\mathcal{C}(\bar{Y}; Z) \times \prod_i^n \mathcal{C}(\bar{X}_i; Y_i) & \xrightarrow{\circ} & \mathcal{C}(\bar{\bar{X}}; Z) \\
F_{\bar{Y}, Z} \times \prod F_{\bar{X}_i, Y_i} \downarrow & & \downarrow F_{\bar{\bar{X}}, Z} \\
\mathcal{D}(F\bar{Y}; FZ) \times \prod_i^n \mathcal{D}(F\bar{X}_i; FY_i) & \xrightarrow{\circ} & \mathcal{D}(F\bar{\bar{X}}; FZ)
\end{array}$$

Definition. Let $F, G : \mathcal{C} \rightarrow \mathcal{D}$ be $\mathcal{V}$ -multifunctors between $\mathcal{V}$ -multicategories. A $\mathcal{V}$ -multinatural transformation $\gamma : F \Rightarrow G$ between them comprises maps

$$\gamma_X : FX \rightarrow GX$$

such that the following diagram commutes:

$$\begin{array}{ccc}
\mathcal{C}(\bar{X}; Y) & \xrightarrow{G_{\bar{X}, Y}} & \mathcal{D}(G\bar{X}; GY) \\
F_{\bar{X}, Y} \downarrow & & \downarrow - \circ \gamma_{\bar{X}} \\
\mathcal{D}(F\bar{X}; FY) & \xrightarrow{\gamma_Y \circ -} & \mathcal{D}(F\bar{X}; GY)
\end{array}$$

Proposition. *There is a (strict) 2-category $\mathcal{V}\text{-MultiCat}$ of $\mathcal{V}$ -multicategories, $\mathcal{V}$ -multifunctors and $\mathcal{V}$ -multinatural transformations.*

Proof. Let $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{E}$ be $\mathcal{V}$ -multifunctors between $\mathcal{V}$ -multicategories. We define their composite $GF : \mathcal{C} \rightarrow \mathcal{E}$ on objects by $(GF)X = G(FX)$ and on hom-objects by the composite map

$$(GF)_{\bar{X}, Y} : \mathcal{C}(\bar{X}; Y) \xrightarrow{F_{\bar{X}, Y}} \mathcal{D}(F\bar{X}; FY) \xrightarrow{G_{F\bar{X}, FY}} \mathcal{E}(GF\bar{X}; GFY).$$

This composition respects identities, since

$$(GF)_{X, X}(1_X) = G_{FX, FX}(F_{X, X}(1_X)) = G_{FX, FX}(1_{FX}) = 1_{GF X}.$$

To see that GF respects the multicategory composition, we can stack the coherence diagrams for F and G as follows:

$$\begin{array}{ccc} \mathcal{C}(\bar{Y}; Z) \times \prod_i^n \mathcal{C}(\bar{X}_i; Y_i) & \xrightarrow{\circ} & \mathcal{C}(\bar{\bar{X}}; Z) \\ \downarrow F_{\bar{Y}, Z} \times \prod F_{\bar{X}_i, Y_i} & & \downarrow F_{\bar{\bar{X}}, Z} \\ \mathcal{D}(F\bar{Y}; FZ) \times \prod_i^n \mathcal{D}(F\bar{X}_i; FY_i) & \xrightarrow{\circ} & \mathcal{D}(F\bar{\bar{X}}; FZ) \\ \downarrow G_{F\bar{Y}, FZ} \times \prod G_{F\bar{X}_i, FY_i} & & \downarrow G_{F\bar{\bar{X}}, FZ} \\ \mathcal{E}(GF\bar{Y}; GFY) \times \prod_i^n \mathcal{E}(GF\bar{X}_i; GFY_i) & \xrightarrow{\circ} & \mathcal{E}(GF\bar{\bar{X}}; GFZ) \end{array}$$

The outer rectangle shows that the composite GF is a valid $\mathcal{V}$ -multifunctor. This composition is associative and unital, with the identity multifunctor sending each object to itself and each hom-object to itself via the identity map.

Now, let $\gamma : F \Rightarrow G$ and $\delta : G \Rightarrow H$ be $\mathcal{V}$ -multinatural transformations. Their composite $\delta \circ \gamma : F \Rightarrow H$ is given by the component maps $(\delta \circ \gamma)_X := \delta_X \circ \gamma_X$. This is a $\mathcal{V}$ -multinatural transformation because γ and δ both are:

$$\begin{array}{ccccc} \mathcal{C}(\bar{X}; Y) & \xrightarrow{H_{\bar{X}, Y}} & \mathcal{D}(H\bar{X}; HY) & & \\ & \searrow G_{\bar{X}, Y} & \downarrow -\circ \delta_{\bar{X}} & & \\ & & \mathcal{D}(G\bar{X}; GY) & \xrightarrow{\delta_Y \circ -} & \mathcal{D}(G\bar{X}; HY) \\ & & \downarrow -\circ \gamma_{\bar{X}} & & \downarrow -\circ \gamma_{\bar{X}} \\ \mathcal{C}(\bar{X}; Y) & \xrightarrow{F_{\bar{X}, Y}} & \mathcal{D}(F\bar{X}; FY) & \xrightarrow{\gamma_Y \circ -} & \mathcal{D}(F\bar{X}; GY) & \xrightarrow{\delta_Y \circ -} & \mathcal{D}(F\bar{X}; HY) \end{array}$$

This composition of transformations is associative and unital. The commutativity of the following diagram verifies the interchange law. [TODO](#)

Therefore $\mathcal{V}$ -multicategories, $\mathcal{V}$ -multifunctors, and $\mathcal{V}$ -multinatural transformations form a strict 2-category. $\square$

Remark. The full sub-2-category of $\mathcal{V}$ -MultiCat comprising $\mathcal{V}$ -multicategories whose only morphisms are unary is isomorphic to $\mathcal{V}$ -Cat.

1.4 Equivalences

Proposition. *The construction of a $\mathcal{V}$ -multicategorical structure on an unbiased monoidal $\mathcal{V}$ -category is the assignment on objects of a 2-functor $M : \mathcal{V}\text{-uMonCat} \rightarrow \mathcal{V}\text{-RepMultiCat}$.*

Proof. Let $\mathcal{C}$ and $\mathcal{D}$ be unbiased monoidal $\mathcal{V}$ -categories, and let $(F, \mu) : \mathcal{C} \rightarrow \mathcal{D}$ be a lax monoidal $\mathcal{V}$ -functor between them. Note that such a map is in particular a $\mathcal{V}$ -functor with components

$$F_{X,Y} : \mathcal{C}(X, Y) \rightarrow \mathcal{D}(FX, FY)$$

for all $X, Y \in \text{ob } \mathcal{C}$.

We need to define $MF : MC \rightarrow MD$ on objects and maps.

On objects, since $\text{ob } MC = \text{ob } \mathcal{C}$, let $MF X := FX$ for all $X \in \text{ob } MC$. On maps, we need assignments of type

$$MF_{\bar{X}, Y} : MC(\bar{X}, Y) \rightarrow MD(MF \bar{X}, MF Y),$$

which is to say of type

$$\mathcal{C}(\bigotimes \bar{X}, Y) \rightarrow \mathcal{D}(\bigotimes F \bar{X}, FY).$$

Thus, define $MF_{\bar{X}, Y}$ to be the composite

$$\mathcal{C}(\bigotimes \bar{X}, Y) \xrightarrow{F_{\bar{X}, Y}} \mathcal{D}(F \bigotimes \bar{X}, FY) \xrightarrow{- \circ \mu} \mathcal{D}(\bigotimes F \bar{X}, FY).$$

We need to check that this definition respects identities and composition. For identities, we have that

$$MF_{X, X}(1_X) := (- \circ \mu_1)F_{X, X}(1_X) = (- \circ \mu_1)1_{FX} = 1_{FX}$$

because μ_1 is an identity. For composition, we need to show that the diagram

$$\begin{array}{ccc} MC(\bar{Y}; Z) \times \prod_i^n MC(\bar{X}_i; Y_i) & \xrightarrow{\quad \circ \quad} & MC(\bar{\bar{X}}; Z) \\ \downarrow MF_{\bar{Y}, Z} \times \prod MF_{\bar{X}_i, Y_i} & & \downarrow MF_{\bar{\bar{X}}, Z} \\ MD(MF \bar{Y}; MF Z) \times \prod_i^n MD(MF \bar{X}_i; MF Y_i) & \xrightarrow{\quad \circ \quad} & MD(MF \bar{\bar{X}}; MF Z) \end{array}$$

commutes. Unravelling the definitions of the vertices and edges of the above diagram, we obtain the following diagram

$$\begin{array}{ccccccc} \mathcal{C}(\bigotimes \bar{Y}, Z) \times \mathcal{C}^n(\bigotimes \bar{X}, \bar{Y}) & \xrightarrow{1 \times \bigotimes F_{\bar{X}, \bar{Y}}} & \mathcal{C}(\bigotimes \bar{Y}, Z) \times \mathcal{C}(\bigotimes \bar{X}, \bigotimes \bar{Y}) & \xrightarrow{\text{comp}} & \mathcal{C}(\bigotimes \bar{X}, Z) & \xrightarrow{- \circ \beta^{-1}} & \mathcal{C}(\bigotimes \bar{X}, Z) \\ \downarrow F_{\bigotimes \bar{Y}, Z} \times F_{\bigotimes \bar{X}, \bar{Y}} & & \downarrow F_{\bigotimes \bar{Y}, Z} \times F_{\bigotimes \bar{X}, \bigotimes \bar{Y}} & & \downarrow F_{\bigotimes \bar{X}, Z} & & \downarrow F_{\bigotimes \bar{X}, Z} \\ \mathcal{D}(F \bigotimes \bar{Y}, FZ) \times \mathcal{D}^n(F \bigotimes \bar{X}, F \bar{Y}) & \xrightarrow{1 \times F_{\bigotimes \bar{X}, \bar{Y}}} & \mathcal{D}(F \bigotimes \bar{Y}, FZ) \times \mathcal{D}(F \bigotimes \bar{X}, F \bigotimes \bar{Y}) & \xrightarrow{\text{comp}} & \mathcal{D}(F \bigotimes \bar{X}, FZ) & \xrightarrow{- \circ F \beta^{-1}} & \mathcal{D}(F \bigotimes \bar{X}, FZ) \\ & \searrow 1 \times \bigotimes F_{\bar{X}, F \bar{Y}} & \downarrow 1 \times (- \circ \mu) & & \downarrow - \circ \mu & & \downarrow - \circ \mu \\ & \mathcal{D}(F \bigotimes \bar{Y}, FZ) \times \mathcal{D}(\bigotimes F \bar{X}, F \bigotimes \bar{Y}) & \xrightarrow{1 \times (\mu \circ -)} & \mathcal{D}(F \bigotimes \bar{Y}, FZ) \times \mathcal{D}(\bigotimes F \bar{X}, F \bigotimes \bar{Y}) & & & \\ & \downarrow (- \circ \mu) \times 1 & & \downarrow (- \circ \mu) \times 1 & & & \\ \mathcal{D}(\bigotimes F \bar{Y}, FZ) \times \mathcal{D}^n(F \bigotimes \bar{X}, F \bar{Y}) & \xrightarrow{1 \times \bigotimes F_{\bar{X}, F \bar{Y}}} & \mathcal{D}(\bigotimes F \bar{Y}, FZ) \times \mathcal{D}(\bigotimes F \bar{X}, F \bar{Y}) & \xrightarrow{\text{comp}} & \mathcal{D}(\bigotimes F \bar{X}, FZ) & & \\ \downarrow 1 \times (- \circ \bar{\mu}) & & \downarrow 1 \times (- \circ \bar{\mu}) & & \downarrow - \circ \bar{\mu} & & \\ \mathcal{D}(\bigotimes F \bar{Y}, FZ) \times \mathcal{D}^n(\bigotimes F \bar{X}, F \bar{Y}) & \xrightarrow{1 \times \bigotimes F_{\bar{X}, F \bar{Y}}} & \mathcal{D}(\bigotimes F \bar{Y}, FZ) \times \mathcal{D}(\bigotimes F \bar{X}, F \bar{Y}) & \xrightarrow{\text{comp}} & \mathcal{D}(\bigotimes F \bar{X}, FZ) & \xrightarrow{- \circ \beta^{-1}} & \mathcal{D}(\bigotimes F \bar{X}, FZ) \end{array}$$

which commutes, being made of naturality and extranaturality squares, along with the coherence for μ . $\square$

Theorem. *The 2-category $\mathcal{V}\text{-uMonCat}$ and the full subcategory of $\mathcal{V}\text{-MultiCat}$ on the representable $\mathcal{V}$ -multicategories are equivalent.*

Proof. We construct a pair of 2-functors between the 2-category of unbiased monoidal $\mathcal{V}$ -categories, $\mathcal{V}\text{-uMonCat}$, and the 2-category of representable $\mathcal{V}$ -multicategories, $\mathcal{V}\text{-RepMultiCat}$, and show they form an equivalence.

First, we construct a 2-functor $M : \mathcal{V}\text{-uMonCat} \rightarrow \mathcal{V}\text{-RepMultiCat}$.

- 0-cells: given an unbiased monoidal $\mathcal{V}$ -category $\mathcal{C}$, let $M\mathcal{C}$ be as defined in the proof that every unbiased monoidal $\mathcal{V}$ -category can be given the structure of a representable $\mathcal{V}$ -multicategory.
- 1-cells: given a lax monoidal $\mathcal{V}$ -functor $F : \mathcal{C} \rightarrow \mathcal{D}$ (which is in particular a $\mathcal{V}$ -functor), let $MF\mathcal{X} := F\mathcal{X}$ and let $MF : M\mathcal{C} \rightarrow M\mathcal{D}$ comprise

$$MF_{\bar{X}, Y} := M\mathcal{C}(\bar{X}; Y) \rightarrow M\mathcal{D}(MF\bar{X}; MFY)$$

defined as the composite

$$\mathcal{C}(\bigotimes \bar{X}, Y) \xrightarrow{F_{\bar{X}, Y}} \mathcal{D}(F \bigotimes \bar{X}, FY) \xrightarrow{- \circ \mu_n} \mathcal{D}(\bigotimes F\bar{X}, FY).$$

This satisfies $MF_{X, X}(1_X) = 1_{MF\mathcal{X}}$ since μ_1 is the identity; the following diagram shows that MF respects composition.

$\square$